

Beyond the Cutoff: Evaluating LLMs for Public Opinion Prediction on Unseen Events

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Abstract

Traditional survey infrastructure for measuring public opinion is eroding: response rates have collapsed, costs are prohibitive for fast-breaking events, and self-selected respondents are increasingly unrepresentative. Large language models (LLMs) have emerged as candidate proxy survey respondents—a practice termed *silicon sampling*—but every prior evaluation tests models on questions fielded *before* their training cutoffs, making it impossible to disentangle genuine distributional reasoning from memorized recall.

We address this gap by evaluating 21 LLMs on OPENPOLL, a benchmark of 92 survey questions paired with temporally anchored Wikipedia event contexts spanning eight real-world events (October 2023–September 2025), all of which post-date every model’s training cutoff. Ground-truth opinion distributions are drawn from contemporaneous YouGov polls. Using a 2×2 factorial design across elicitation channel (next-token probability vs. verbalized distribution) and context (zero-shot vs. few-shot), we find that elicitation mode is a primary driver of accuracy: few-shot next-token prompting yields the best aggregate performance (mean Wasserstein Distance 0.158), and its benefit is asymmetric—substantially improving base models while leaving fine-tuned models neutral or degraded. Opinion-domain and alignment fine-tuning do not generalize to out-of-distribution events; fine-tuned models underperform their base counterparts in most conditions, with the sole exception of SubPop fine-tuning on Mistral-7B. Finally, inter-model disagreement is a *negative* predictor of human opinion entropy, reinterpreting it as a signal of model-level prediction difficulty rather than population-level contestedness. Together, these results surface key boundary conditions on when LLM-based opinion prediction can and cannot be trusted.

Keywords

Large Language Models, Public Opinion Prediction, Computational Social Science

1 Introduction

Public opinion is a fundamental signal in democratic societies. It provides insight into the long-term stability of political systems, informs policy decisions, and serves as a check on institutional power [4]. Yet the infrastructure for measuring public opinion is eroding. Response rates from telephone surveys have collapsed from roughly 36% in 1997 to approximately 6% today [19], and even professionally administered arrangements like YouGov surveys can cost thousands of dollars and take days to field [4]. This cost is

particularly acute for novel, fast-breaking events, where the window for timely opinion measurement may close before a survey can be deployed [29]. Compounding this, surveys that do complete are increasingly unrepresentative: those who respond tend to be more politically engaged, distorting the very signal the surveys are meant to capture [9, 10].

Against this backdrop, large language models (LLMs) have emerged as a candidate technology for *silicon sampling* [2]—the practice of using LLMs as proxy survey respondents to estimate public opinion distributions at a fraction of the cost and time of traditional polling. Prior work has shown that LLMs conditioned on sociodemographic backstories can reproduce subgroup opinion distributions from established surveys with surprising fidelity [2, 36]. However, every prior evaluation in this space has a fundamental methodological limitation: the survey questions used to evaluate models were fielded *before* the models’ training cutoffs. This renders it impossible to distinguish genuine distributional reasoning from memorized recall of opinion data the model absorbed during pretraining [15, 37].

This paper asks a different question. Rather than asking whether LLMs can predict opinion, we ask *when their predictions should be trusted at all*, specifically for novel events that fall outside any model’s training window—the out-of-distribution (OOD) setting where the stakes of reliable estimation are highest and ground truth is hardest to obtain.

We make three contributions toward answering this question.

The OPENPOLL dataset. We introduce OPENPOLL, a benchmark of ninety-two survey questions paired with temporally anchored event descriptions, spanning eight real-world events from October 2023 to September 2025—all of which postdate the training cutoffs of every model evaluated. Ground-truth opinion distributions are drawn from YouGov polls fielded at the time of each event. Event context is constructed using Wikipedia’s MediaWiki API revision history, retrieving only the article content that existed on or before each survey’s fielding date. This temporal non-contamination guarantee ensures that model predictions cannot rely on retrospective information, providing a principled OOD evaluation setting. The dataset spans both political events (the October 7th, 2023 attacks, the Fall of the Assad regime, the Biden 2024 withdrawal, the Charlie Kirk assassination) and non-political events (the Drake–Kendrick Lamar feud, the Taylor Swift–Travis Kelce engagement, the trial of Sean Combs), enabling analysis of whether model capabilities generalize across domains.

A systematic elicitation mode comparison. We evaluate 21 models—spanning base, instruction-tuned, subpopulation fine-tuned (SUBPOP), and alignment fine-tuned (SOCRATES SFT/DPO) variants—across a

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2×2 factorial design of elicitation channel (next-token probability NTP vs. verbalized distribution VD) and context (zero-shot vs. few-shot). Every prior silicon sampling study has committed to a single elicitation mode, making any interaction between prompting mode and model class structurally invisible. Our design makes it visible. We find a consistent aggregate ordering (FS-NTP < FS-VD < VD < NTP in mean Wasserstein Distance), with few-shot context providing substantially larger gains for next-token elicitation ($\Delta\text{WD} = 0.060$) than for verbalized elicitation ($\Delta\text{WD} = 0.025$). Critically, this benefit is asymmetric across model classes: few-shot next-token prompting consistently improves base models while leaving fine-tuned models neutral or degraded, suggesting that fine-tuning encodes distributional priors that in-context demonstrations cannot improve upon—and may disrupt.

Fine-tuning does not generalize to OOD events. Across the 21-model cohort, opinion-domain and alignment fine-tuning does not reliably improve OOD opinion prediction. SubPop fine-tuning on the Mistral base is the sole exception, achieving a modest mean improvement ($\Delta\text{WD} = -0.022$) with consistent win rates across both political and non-political events. The Socrates SFT models, by contrast, exhibit sharp degradation—socrates-llama3-8b-sft degrades by +0.160 WD relative to its base, the largest delta in the cohort, attributable to a chat-template format mismatch that causes the model to fall back to near-uniform output distributions. The DPO variants are more robust to prompt format variation but still underperform their base models in most conditions. These results are consistent with classical distributional-shift overfitting: fine-tuning increases a model’s fit to in-distribution survey patterns at the cost of generalization to novel events.

Inter-model disagreement as a label-free reliability signal. We introduce and evaluate $\Delta^{(e)}(q)$, the inter-model standard deviation across the twenty-one-model ensemble, as a label-free proxy for prediction reliability computable without any access to ground-truth polling data. Drawing on the deep-ensembles literature [23], we hypothesize that high inter-model disagreement should correlate with questions on which human opinion is genuinely fragmented (high normalized entropy). The empirical results disconfirm this hypothesis: disagreement is negatively correlated with human entropy. We attribute this to a ceiling effect—when human opinion is highly diffuse, all models are mechanically pushed toward near-uniform output, suppressing ensemble variance. Conversely, peaked human distributions present harder allocation problems on which models diverge. Reinterpreted, $\Delta^{(e)}(q)$ flags model-level prediction difficulty rather than population-level opinion contestedness. Flagging high-disagreement questions for additional scrutiny or ensemble averaging remains actionable even under this reinterpretation, as it identifies polls that are structurally hard for LLMs to estimate reliably.

Research Questions. This paper addresses the following research questions:

RQ1 *How do LLMs perform on out-of-distribution public opinion prediction, and how does elicitation mode modulate that performance?*

RQ2 *Does opinion-domain or alignment fine-tuning improve OOD opinion prediction, and does the answer differ across fine-tuning strategies?*

RQ3 *Does inter-model disagreement serve as a reliable label-free signal for identifying unreliable opinion estimates?*

Contributions. In summary, this paper makes the following contributions:

- OPENPOLL, a temporally grounded, contamination-resistant benchmark of ninety-two OOD public opinion questions paired with Wikipedia-anchored event contexts and YouGov ground-truth distributions.
- A 21-model $\times$ 4-mode evaluation revealing that elicitation design is a primary driver of predictive accuracy, and that its effect is model-class dependent in ways invisible to single-mode evaluations.
- Evidence that opinion-domain and alignment fine-tuning do not reliably generalize to OOD events, with fine-tuned models underperforming their base counterparts in most conditions.
- An empirical reinterpretation of inter-model disagreement as a proxy for model-level prediction difficulty rather than population-level opinion contestedness, with practical implications for when to trust LLM opinion estimates.

2 Related Work

2.1 Prior Work

Silicon sampling and LLMs as survey respondents. A growing body of research has explored whether large language models can serve as proxies for human survey respondents, a paradigm Argyle et al. [2] term *silicon sampling*. Their foundational study showed that GPT-3, conditioned on sociodemographic backstories, can reproduce subgroup opinion distributions from the American National Election Study with surprising fidelity, introducing *algorithmic fidelity* as a measure of how well a simulated subpopulation mirrors its human counterpart. Santurkar et al. [36] extended this inquiry to nine models via the OpinionsQA benchmark, constructed from Pew Research surveys, finding pervasive misalignment between default model responses and U.S. public opinion, measuring a gap as large as the partisan divide on climate change, and only partly remedied by demographic steering. Bisbee et al. [5] reached similarly cautionary conclusions: while GPT-3.5 replicates mean ANES feeling-thermometer scores, it collapses response variance, yields unstable regression coefficients, and exhibits marked sensitivity to prompt wording. Together, this thread establishes that LLMs can partially capture population-level opinion structure but fail systematically in ways that have not yet been characterized by a unifying diagnostic.

Political bias and ideological skew. Parallel work has characterized the ideological dispositions LLMs exhibit by default. Hartmann et al. [16] applied standard political orientation instruments to ChatGPT across multiple languages, finding consistent left libertarian skew. Feng et al. [13] traced this bias along the full pipeline—from pretraining corpora through model weights to downstream task performance—demonstrating that partisan signals in text corpora translate into predictable asymmetries in model outputs. Tjauatja

et al. [42] further showed that instruction-tuned models fail to mirror known human survey biases (e.g., acquiescence effects) and are disproportionately sensitive to prompt perturbations that leave human respondents unaffected, complicating the use of persona conditioning as a corrective. Durmus et al. [12] broadened the audit globally via the GlobalOpinionQA dataset, revealing a default Western bias that persists even under country-perspective prompting. Critically, none of these studies offer practitioners a way to know, in advance, which questions are most likely to produce unreliable LLM estimates.

Benchmark contamination and evaluation validity. The reliability of LLM evaluation is further complicated by data contamination. Golchin and Surdeanu [15] demonstrated that widely used benchmarks are already memorized by frontier models, proposing guided-instruction probes that detect contamination with 92–100% accuracy without access to training data. Shi et al. [37] introduced WikiMIA, a temporally stratified contamination benchmark exploiting Wikipedia’s event chronology relative to model training cutoffs, providing a principled design template for contamination-resistant evaluation. Sainz et al. [35] argued that the NLP evaluation paradigm itself is compromised when benchmarks circulate publicly before models are trained, and Dekoninck et al. [11] showed that existing detection methods are trivially evaded through paraphrasing, inflating apparent performance by up to 15%. Magar and Schwartz [25] drew the foundational distinction between mere memorization and active exploitation of contaminated data, establishing that practical consequences depend on factors beyond test-set overlap alone. These findings collectively motivate not just better detection of contamination, but contamination-resistant *benchmark design* as a first-class methodological goal. The HELM framework [24] represents the most ambitious attempt at holistic LLM evaluation to date; OPENPOLL addresses the gap it leaves in opinion-prediction tasks where temporal contamination is a first order concern.

Computational social science and opinion dynamics. Within computational social science, NLP approaches to political opinion have largely focused on *stance detection*, looking at how models classify individual documents as pro or con for a given target. Augenstein et al. [3] established bidirectional conditional encoding as the canonical architecture, and Mohammad et al. [28] formalized the task through the SemEval-2016 shared task on tweet-level stance classification. This document centric framing differs fundamentally from the population level opinion distribution prediction studied here. From the political science side, Broockman and Kalla [6] and Wojcieszak [47] demonstrate that attitude change is most pronounced precisely where population level opinion is both consequential and hardest to measure reliably, that being on contested social issues. More recently, Parikh et al. [33] tracked nine LLM configurations daily during the 2024 U.S. presidential election, finding systematic 10–40% favorability overprediction and high cross-model variance, providing the most directly comparable multi-model analysis of LLM opinion accuracy to date.

Uncertainty estimation in deep learning and LLMs. Lakshminarayanan et al. [23] established that ensemble variance across independently trained networks constitutes the most reliable proxy for epistemic

uncertainty in deep learning, outperforming single-model calibration. Kadavath et al. [18] showed that frontier LLMs are well-calibrated when self-reporting confidence on multiple-choice items, while Kuhn et al. [22] extended uncertainty quantification to generation tasks by clustering semantically equivalent outputs. Kruse et al. [21] most recently proposed MUSE, an information-theoretic framework using Jensen-Shannon divergence across multiple heterogeneous LLMs to aggregate uncertainty signals without access to model internals. Methodologically, these works anchor the intuition that inter-model disagreement captures something real about predictive uncertainty—an intuition this paper extends to a domain where the signal carries an interpretable semantic meaning. Tang et al. [39] extend this line of inquiry to the cross-lingual setting, finding that LLM probability estimates for natural language hedges (e.g., “likely”, “almost certain”) diverge from human interpretations in ways that vary systematically by language and gendered context, providing independent evidence that verbalized uncertainty expressions in LLMs are not reliably calibrated to human compatible probability scales.

Fine-tuning and alignment methods. The fine-tuning diagnostics in this paper build on two foundational alignment methods: RLHF-based supervised fine-tuning (SFT), formalized at scale by Ouyang et al. [31] via InstructGPT, and Direct Preference Optimization [34], which reframes alignment as an implicit reward-learning problem without an explicit reward model. Both methods alter the prior over outputs in ways that affect opinion-elicitation behavior, motivating a systematic comparison against subpopulation fine-tuning as an alternative adaptation strategy.

MCQ evaluation reliability, selection bias, and verbalized calibration. A parallel methodological thread has examined whether first-token probability extraction, which exists as the most common MCQ evaluation protocol, successfully captures what instruction-tuned models actually believe. Zheng et al. [51] conduct an extensive empirical study across 20 LLMs on three benchmarks and find that models exhibit ubiquitous selection bias, in that they assign systematically higher prior probability mass to specific option-ID tokens regardless of question content, a token-level artifact that persists across model families and cannot be mitigated by chain-of-thought prompting or explicit debiasing instructions. In their zero-shot MMLU evaluation, simply relocating the correct answer to position D degrades gpt-3.5-turbo accuracy by 6.3 points. Wang et al. [45] further show that for instruction-tuned models, first-token probabilities diverge from the model’s own text-form answer on the same question at mismatch rates exceeding 60% on OpinionQA, driven by conversational preambles (e.g., “Sure!”) and safety refusals that displace probability mass from the option tokens. The divergence is most severe for models heavily fine-tuned on conversational or safety data, which is precisely the class of models that dominate the literature on opinion elicitation and our own Tier 1 cohort.

Crucially, Tian et al. [41] provide a constructive complement to this, finding that while RLHF systematically worsens the calibration of conditional log-probabilities, the verbalized confidence expressions emitted by those same RLHF-trained models are typically *better*-calibrated than their log-probability counterparts across

TriviaQA, SciQ, and TruthfulQA, often reducing expected calibration error by a relative 50%, without any fine-tuning for verbalization. This asymmetry—RLHF degrades the implicit channel while partially recovering calibration through the explicit one—directly motivates treating NTP and VD as measuring fundamentally different quantities rather than interchangeable approximations of the same underlying belief state, and provides the theoretical grounding for why a verbalized few-shot condition may close performance gaps that zero-shot NTP cannot.

In-context learning and few-shot evaluation. The few-shot elicitation modes in this study build on the in-context learning paradigm introduced by Brown et al. [7], who showed that large language models can adapt to new tasks from a small number of input–output demonstrations prepended to the prompt, without any gradient update. Subsequent work has sharpened understanding of why this works. Min et al. [26] find that randomly replacing demonstration labels with incorrect ones barely degrades performance across a range of classification and multiple-choice tasks, suggesting that demonstrations primarily teach the model the label space, input distribution, and output format rather than the correct input–output mapping itself. In the opinion-prediction setting this finding leads to the implication that if demonstrations teach format rather than content, a few-shot verbalized-distribution mode that supplies ground-truth survey distributions as demonstration answers carries qualitatively richer signal than standard ICL, because the distributional label itself is the object of interest. This distinction motivates the separation between FS-NTP—where demonstrations encode only the modal option—and FS-VD—where demonstrations encode the full human distribution—as probes of whether models exploit distributional signal or merely format.

Few-shot prompting is not, however, free of its own pathologies. Zhao et al. [50] identify three systematic biases in few-shot inference, majority label bias (the model is drawn toward whichever label appears most frequently in the demonstrations), recency bias (the model is drawn toward whichever label appears last), and common-token bias (the model prefers tokens that are frequent in its pretraining data). All three can cause the model to anchor on surface statistics of the demonstration set rather than its semantic content, which in the opinion-prediction context means echoing distributional features of the final demonstration rather than reasoning about the target question. We treat these as structural constraints on the interpretation of few-shot results rather than engineering problems to be solved, and hold k and the random seed fixed across all models to ensure that any observed differences across tiers and modes are not confounded by demonstration selection variance.

2.2 Positioning This Work

The body of work reviewed above represents genuine, substantial progress. Silicon sampling methods have shown that LLM opinion estimates are nontrivially informative, contamination researchers have developed rigorous tools for detecting evaluation failures, and the CSS community has built strong task definitions and benchmarks for political NLP. Our work does not dispute these contributions, it instead asks a different, complementary question. Not whether LLMs can predict opinion, but when their predictions should be trusted at all when it comes to new events.

Three gaps in the existing literature motivate our approach. First, every prior silicon sampling evaluation Argyle et al. [2], Santurkar et al. [36], Bisbee et al. [5], and related work, all use opinion questions from surveys that predate model training cutoffs. The contamination literature makes clear that this conflates genuine reasoning with memorized recall. OPENPOLL addresses this by grounding all 98 evaluation items in post cutoff events, using Wikipedia’s temporal structure [following 37] to provide reproducible factual context while ensuring the events themselves fall outside any model’s training window.

Second, prior work has not offered practitioners a label free uncertainty signal, a way to identify, without access to ground truth polling data, which LLM opinion estimates are most likely to be unreliable. Existing self calibration approaches [18] reflect confidence within a model, not an inter-model contestation. We show that ensemble standard deviation across 21 heterogeneous models—an application of the deep-ensemble intuition from Lakshminarayanan et al. [23] to a new domain—correlates with ground-truth question difficulty in a way that is interpretable: high-disagreement questions systematically map to politically contested topics where even humans hold divided views.

Third, the literature has largely treated elicitation method as a nuisance variable to be controlled rather than an informative signal about model behavior. The MCQ evaluation reliability work reviewed above [41, 45, 51] establishes that the choice of elicitation channel is not merely a procedural detail, it actively modulates which aspect of model behavior is measured, and its effect depends on the fine-tuning regime. The in-context learning literature [7, 26, 50] adds a further dimension. Whether and how models exploit distributional exemplars depends on what demonstrations actually teach, which may differ from what the experimenter intends. Yet no prior opinion-prediction study has systematically varied elicitation mode, much less crossed it with in-context context, across a controlled model cohort to measure the interaction. Our 21-model $\times$ 4-mode design, spanning the 2×2 factorial of channel (NTP vs. VD) and context (zero-shot vs. few-shot), reveals interactions invisible to any single-mode evaluation. Next-token probability elicitation outperforms zero-shot verbalization for subpopulation fine-tuned models, while the advantage reverses for instruction-tuned base models, and the few-shot conditions diagnose whether that gap reflects positional bias or a deeper failure of distributional reasoning. This interaction has direct implications for benchmark design—the choice of elicitation mode should be model-class-aware rather than uniform—and for fine-tuning diagnostics, where it helps explain the performance gap between alignment-based (SFT/DPO) and subpopulation-based adaptation strategies.

Taken together, OPENPOLL, the disagreement metric, and the prompting-mode analysis constitute a methodological contribution to evaluation practice, not merely a new leaderboard. The paper’s primary claim is not that any model achieves low error rate on public opinion prediction—it is that inter-model disagreement is a reliable, label-free signal for deciding when *not* to trust the estimate.

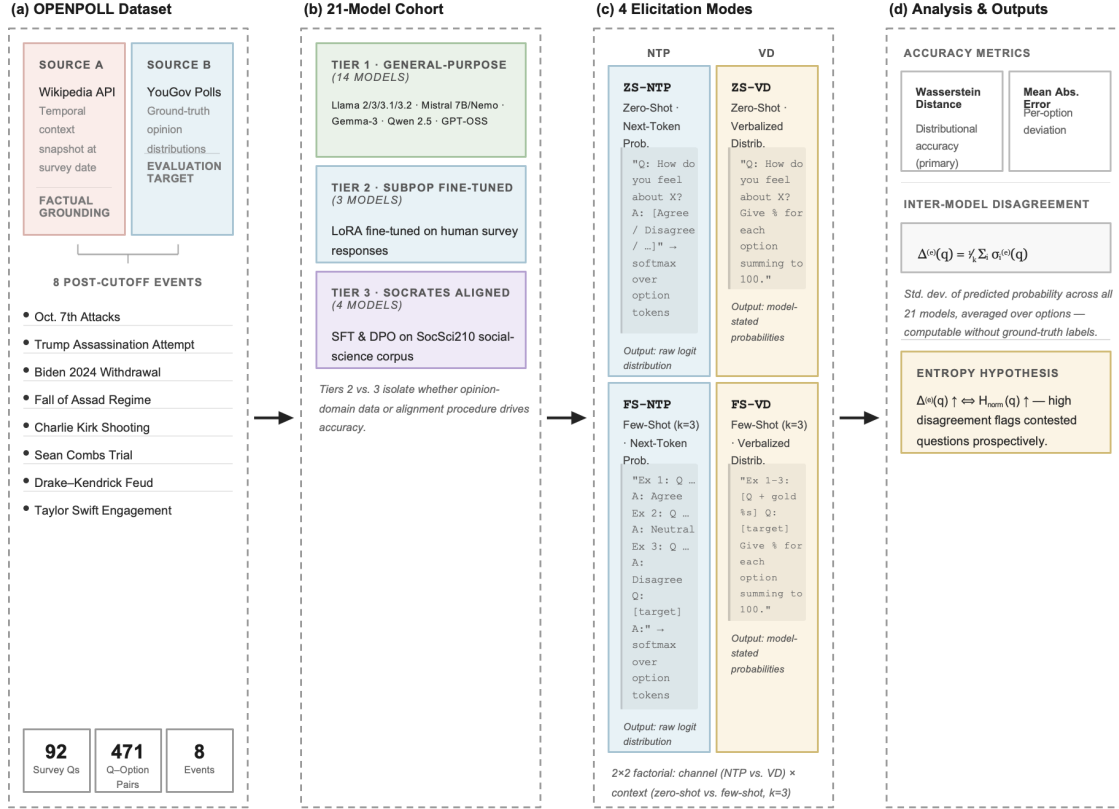


Figure 1: Our Method workflow; we collect data from YouGov Polls, and scrape Temporal Event Context from the associated Wiki page, test the models across the 4 elicitation modes, then evaluate the resulting performance

3 Methodology

3.1 Constructing OPENPOLL

OPENPOLL is a dataset pairing human opinion distributions with temporally anchored event descriptions for a set of real world events that span 2023–2025. It draws on two primary sources, structured survey data capturing ground-truth public opinion and event descriptions derived from Wikipedia’s revision history.

Survey Data. The ground truth opinion distributions are drawn from YouGov polls conducted in close proximity to each event. The dataset covers eight events: the October 7th, 2023 attacks, the attempted assassination of Donald Trump in Pennsylvania, the Fall of the Assad regime, the Joe Biden 2024 withdrawal, the Charlie Kirk assassination, the trial of Sean Combs, the Drake–Kendrick Lamar feud, and the Taylor Swift - Travis Kelce Engagement. For each event, multiple survey questions are included, yielding 471 question–response–option pairs in total. Each record includes the full probability distribution over response options for the overall adult population, and some as well include breakdowns across demographic subgroups (region, gender, political affiliation, age, and race).

Temporal Anchoring via Wikipedia Revision History. To provide models with an event description that reflects only the information

available at the time a given survey was fielded, we construct event contexts using Wikipedia’s MediaWiki API rather than static article snapshots. For each question–event pair, we use the survey fielding date as a cutoff and query the API for the most recent revision of the corresponding Wikipedia article that existed on or before that date. We then fetch the rendered HTML for that specific revision and extract its lead paragraph using BeautifulSoup, stripping infoboxes, hatnotes, tables of contents, and citation superscripts. The extracted revision ID and timestamp are retained as metadata, enabling exact reconstruction of every input context.

This approach ensures temporal non-contamination. Essentially, the event description seen by a model corresponds to what Wikipedia contained at the moment the corresponding survey was conducted, not a later, retrospectively enriched version of the article. We use the Wikipedia API directly rather than an external archival service such as the Internet Archive Wayback Machine avoiding the overhead of snapshot availability and timestamp mismatches that complicate Wayback based pipelines, while still grounding each context in a verifiable historical revision.

We use wikipedia to provide this temporal anchor due to the standardization it provides as well as bias concerns that would be more likely to be present if using other sources

Table 1: Normalized entropy of opinion distributions across events.

Event	Normalized Entropy
Drake Vs Kendrick	0.964837
October 7th, 2023	0.960812
Charlie Kirk Shooting	0.959691
Assassination attempt on Donald Trump	0.931165
Joe Biden 2024 Drop Out	0.870782
Fall of Assad Regime	0.863006
Taylor Swift Engagement	0.805240
Trial of Sean "Diddy" Combs	0.680375

Dataset Structure. The final dataset is stored as a structured JSON file in which each entry maps a survey question to its associated Wikipedia metadata (canonical URL, revision ID, revision timestamp, API endpoint, and extracted lead text) alongside the response options and their empirical probabilities. This format allows any downstream evaluation to reconstruct the exact information context under which each question was posed, and to pair that context directly with the human distribution it should be compared against.

Dataset Summary. The dataset is made up of 92 questions across the 8 events covering events from October 2023 - September 2025. The dataset is dominated by political events, which exists as a consequence of what is actively polled by companies like YouGov. To understand if this leads to a bias such that political events are over-represented and over understood by the models, we separate the political and non-political events under some analyses. To understand the makeup of each event’s distributions, We calculate the Shannon entropy normalized at the event level. We use entropy measures how evenly spread public opinion is across the response options for a given survey question. It quantifies whether a question has a clear majority view or whether opinion is fragmented across options across the human populations.

We define normalized Shannon entropy as

$$H_{\text{norm}} = \frac{H}{\ln(n)} = \frac{-\sum_{i=1}^n p_i \ln(p_i)}{\ln(n)}$$

where p_i is the proportion of survey respondents choosing response option i and n is the number of response options for that question. The normalization factor $\ln(n)$ represents the maximum possible entropy, achieved when opinion is perfectly uniform across all options (i.e., $p_i = \frac{1}{n}$ for all i). As a result, $H_{\text{norm}} \in [0, 1]$, where 0 indicates complete consensus (all respondents select the same option) and 1 indicates maximally contested opinion (respondents are evenly split across all options). Normalizing by $\ln(n)$ is necessary because questions in the dataset have varying numbers of response options (ranging from 3 to 12); without normalization, questions with more options would mechanically yield higher entropy, confounding differences in the number of options with genuine differences in opinion contestedness.

Table 1 summarizes the distribution of H_{norm} across events and question topics for the US population. Entropy values across all events range from 0 to 1.665, with a mean of 0.913 ($SD = 0.192$),

indicating that the dataset captures generally unbalanced distributions, but still cover a broad spectrum of opinion contestedness. We use H_{norm} throughout the analysis as a means to understand model disagreement.

3.2 Evaluation Cohort and Experimental Design

The 20 models evaluated in this study were not assembled by convenience but according to a deliberate three-tier design whose structure is the experimental precondition for addressing Gap 3 from Section 2. The first tier comprises general purpose LLMs [1, 27, 40, 49] that have received no opinion domain training of any kind. These models serve as the population level baseline against which finetuning effects are isolated. The second tier introduces subpopulation finetuning via the SUBPOP family [38]. These models have been finetuned directly on human survey response data and optimized to reproduce observed demographic opinion distributions. The third tier introduces alignment finetuning via the SOCRATES family [20], featuring models adapted through supervised finetuning (SFT) [31] and direct preference optimization (DPO) [34] on a social science questionnaire corpus. The distinction between Tiers 2 and 3 is the locus of the Gap 3 question: without both, it is impossible to determine whether improvements in opinion prediction accuracy arise from exposure to opinion structured data specifically, or from general alignment to human preferences more broadly.

Tier 1: General-Purpose Base LLMs. The baseline tier spans 13 models drawn from four widely-deployed open-weight families. From the Llama family we include Llama-2-7B and Llama-2-13B (base weights) [43], Meta-Llama-3-8B (base), Llama-3.1-8B-Instruct, Llama-3.2-1B-Instruct, and Llama-3.2-3B-Instruct [1]. From the Mistral family we include Mistral-7B-v0.1 (base), Mistral-7B-Instruct-v0.3, and Mistral-Nemo-Instruct-2407 [?]. The Gemma-3 family contributes three instruction-tuned variants: 270M-it, 4B-it, and 12B-it [40]. The tier is completed by Qwen2.5-14B [49] and openai/gpt-oss-20b [30]. Model sizes range from 270 million to 20 billion parameters, providing coverage across the open-weight regime. Where base and instruction-tuned variants of the same architecture are both present (e.g., Mistral-7B-v0.1 alongside Mistral-7B-Instruct-v0.3), the pairing enables within-family comparisons of the effect of general instruction following on opinion-elicitation behavior, independent of any opinion-domain training.

Tier 2: Subpopulation Fine-Tuned Models (SUBPOP Family). The second tier consists of three models from the SUBPOP family [38]: jjssuh/llama-2-7b-subpop, jjssuh/llama-2-13b-subpop, and jjssuh/mistral-7b-v0.1-subpop. Each is a LORA [17] finetune of its Tier-1 base counterpart trained on human survey response data to reproduce the opinion distributions observed across demographic subgroups. These models constitute the strongest prior work baseline for the opinion prediction task: they have seen opinion structured training data and are therefore the models most likely to perform well on OPENPOLL by design. Their inclusion allows a controlled comparison of opinion domain finetuning effects against both the base models from which they were derived and the alignment finetuned models in Tier 3.

Tier 3: Alignment Fine-Tuned Models (SOCRATES Family). The third tier consists of four models from the SOCRATES family [20]: socrates-llama3-8b-sft, socrates-llama3-8b-dpo, socrates-qwen2.5-14b-sft, and socrates-qwen2.5-14b-dpo. These models were finetuned on the SocSci210 dataset, a corpus of social science questionnaire items, using SFT and DPO respectively. Unlike the SUBPOP models, which learn to mimic empirical survey distributions, SOCRATES models are trained to respond appropriately to survey style prompts under alignment objectives that do not directly supervise on population level outcomes. The SFT and DPO variants are included together to assess whether the alignment procedure itself, not merely the domain of fine-tuning, modulates opinion-prediction behavior. Paired with Tier 2, Tier 3 makes the Gap 3 comparison concrete: fine-tuning toward observed opinion data versus fine-tuning toward human preference alignment, evaluated identically on OPENPOLL.

Table 2 provides a complete enumeration of all 20 models with their tier assignments, base architectures, adaptation methods, and approximate parameter counts.

3.3 Opinion Elicitation Methods

This study evaluates four opinion elicitation modes arranged in a 2×2 factorial design, varying two factors, the elicitation channel (next-token probability NTP vs. verbalized distribution VD) and *context* (zero-shot vs. few-shot). Each mode represents a distinct combination of these factors, making them the independent variable of the experiment rather than arbitrary implementation choices included for completeness. Each cell of this design is a distinct mapping from a model and a survey question to a probability distribution over response options, and the four mappings differ in their mechanics, their prompt structure, and, critically, in the theoretical assumptions under which each should be preferred. Holding the 21-model cohort and the OPENPOLL items fixed while varying elicitation mode is what makes the mode-by-model-tier interaction in Section 5 visible at all; every prior study in this literature has used a single mode per paper, rendering that interaction invisible by design.

Mode 1: Zero-Shot Next-Token Probability (NTP). The first mode extracts the model’s opinion distribution directly from its next-token probability mass. For a survey question q with response options $\mathcal{R} = \{r_1, \dots, r_k\}$, the prompt presents q followed by an answer stem (e.g., “Answer:”), and the distribution is constructed from the model’s conditional log-probabilities over the leading token of each option:

$$\hat{p}_i^{\text{NTP}} = \frac{\exp(\text{logit}(r_i | q))}{\sum_{j=1}^k \exp(\text{logit}(r_j | q))}, \quad (1)$$

where $\text{logit}(r_i | q)$ is the raw log-probability assigned to the first token of option r_i given the prompt context, and the denominator renormalizes over the constrained option set. No generation is required; a single forward pass with a vocabulary lookup suffices.

NTP Prompt Structure

```
Context: {wikipedia_lead}

Survey question: {question_text}

Options:
A. {option_1}
B. {option_2}
...

Answer:
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The theoretical case for NTP rests on access and economy. The next-token distribution is the model’s implicit belief state. It pulls the distribution it must have computed to generate any continuation at all and is accessible without requiring the model to engage in any additional reasoning or self-report. For base (non-instruction-tuned) models, NTP is often the only viable zero shot mode: asking such a model to “provide a percentage breakdown” risks producing a malformed output, while its token probabilities remain well defined. More broadly, NTP is the lowest cost mode and the one least susceptible to self presentation effects introduced by RLHF alignment, since the distribution is read off before any sampling or alignment shaped decoding policy is applied.

Distribution extraction requires no generation; a single forward pass suffices and temperature is undefined for this mode.

These theoretical advantages must be weighed against two known empirical pathologies. Zheng et al. [51] show that LLMs exhibit pervasive selection bias, which presents itself as a systematic prior over option-ID tokens (e.g., always favoring A or C) that cannot be removed by chain-of-thought or debiasing instructions. Wang et al. [45] demonstrate that for instruction-tuned models, first-token probabilities diverge from the model’s own text-form answer at mismatch rates exceeding 60%, driven by alignment-induced preambles and safety refusals on sensitive topics. Both pathologies are structural rather than incidental, motivating the other modes explored in this study.

Mode 2: Zero-Shot Verbalized Distribution (VD). The second mode asks the model to make its beliefs explicit by verbalizing a probability distribution over the response options. The model is prompted to assign fractional probabilities to each option such that the values sum to 1, and the output is parsed into a discrete distribution $\hat{p}^{\text{VD}}$.

Table 2: Model groupings

Model ID	Tier	Base Architecture	Adaptation Method	Approx. Params
google/gemma-3-270m-it	1 – General-Purpose	Gemma 3	Instruction tuning (SFT+RLHF)	270M
meta-llama/Llama-3.2-1B-Instruct	1 – General-Purpose	Llama 3.2	Instruction tuning (RLHF)	1B
meta-llama/Llama-3.2-3B-Instruct	1 – General-Purpose	Llama 3.2	Instruction tuning (RLHF)	3B
google/gemma-3-4b-it	1 – General-Purpose	Gemma 3	Instruction tuning (SFT+RLHF)	4B
meta-llama/Llama-2-7b-hf	1 – General-Purpose	Llama 2	None (base weights)	7B
mistralai/Mistral-7B-v0.1	1 – General-Purpose	Mistral	None (base weights)	7B
mistralai/Mistral-7B-Instruct-v0.3	1 – General-Purpose	Mistral	Instruction tuning (SFT+DPO)	7B
meta-llama/Meta-Llama-3-8B	1 – General-Purpose	Llama 3	None (base weights)	8B
meta-llama/Llama-3.1-8B-Instruct	1 – General-Purpose	Llama 3.1	Instruction tuning (RLHF)	8B
mistralai/Mistral-Nemo-Instruct-2407	1 – General-Purpose	Mistral Nemo	Instruction tuning (SFT+DPO)	12B
google/gemma-3-12b-it	1 – General-Purpose	Gemma 3	Instruction tuning (SFT+RLHF)	12B
meta-llama/Llama-2-13b-hf	1 – General-Purpose	Llama 2	None (base weights)	13B
Qwen/Qwen2.5-14B	1 – General-Purpose	Qwen 2.5	None (base weights)	14B
openai/gpt-oss-20b	1 – General-Purpose	GPT-OSS	Undisclosed	20B
jjssuh/llama-2-7b-subpop	2 – SubPop Fine-Tuned	Llama 2	LORA on survey response data	7B
jjssuh-mistral-7b-v0.1-subpop	2 – SubPop Fine-Tuned	Mistral	LORA on survey response data	7B
jjssuh/llama-2-13b-subpop	2 – SubPop Fine-Tuned	Llama 2	LORA on survey response data	13B
socratesft/socrates-llama3-8b-sft	3 – Socrates Fine-Tuned	Llama 3	SFT on SocSci210	8B
socratesft/socrates-llama3-8b-dpo	3 – Socrates Fine-Tuned	Llama 3	DPO on SocSci210	8B
socratesft/socrates-qwen2.5-14b-sft	3 – Socrates Fine-Tuned	Qwen 2.5	SFT on SocSci210	14B
socratesft/socrates-qwen2.5-14b-dpo	3 – Socrates Fine-Tuned	Qwen 2.5	DPO on SocSci210	14B

VD Prompt Structure

```
Context: {wikipedia_lead}

Survey question: {question_text}

Estimate the percentage of U.S. adults who would
choose each option.
Your estimates must sum to 1.

Options:
A. {option_1}
B. {option_2}
...

Respond with only a JSON object of the form
{"A": <float>, "B": <float>, ...}.
```

Verbalized outputs are generated with greedy decoding ($T = 0$) for deterministic, reproducible distributions; full hyperparameter specification is in Section 4.1.

VD has two theoretical advantages over NTP. First, it makes the model’s reasoning available for inspection by forcing the model to produce a coherent distribution rather than merely assigning a logit to a token. Second, VD is structurally compatible with in-context learning because its prompt elicits structured output—question–distribution pairs from other events can be prepended as demonstrations, as exploited in Mode 4 below.

The theoretical cost of VD is that instruction following training shapes the model’s verbal self reports in ways that may depart

from the implicit distribution captured by NTP. Xiong et al. [48] find that LLMs are pervasively overconfident when verbalizing confidence under zero-shot prompting. A further structural concern identified by Wang et al. [44] is that verbalized distributions cannot be straightforwardly calibrated by standard post-hoc methods. For example, temperature scaling assumes unnormalized logits as input rather than an already normalized distribution, so applying it to verbalized probabilities causes a re-softmax distortion. We do not apply post-hoc calibration to VD outputs for this reason, treating the raw verbalized distribution as the elicited signal.

Mode 3: Few-Shot Next-Token Probability (FS-NTP). The third mode prepends k demonstration turns before extracting next-token probabilities, isolating whether the positional and selection biases that afflict zero-shot NTP can be suppressed through in-context examples. Each demonstration pairs a survey question with its modal response (the option letter corresponding to the highest empirical probability in the YouGov data) providing a minimal but contamination safe in-context signal:

$$\hat{p}_i^{\text{FS-NTP}} = \frac{\exp(\text{logit}(r_i | q, \{(q_j^*, \ell_j^*)\}_{j=1}^k))}{\sum_{m=1}^{|\mathcal{R}|} \exp(\text{logit}(r_m | q, \{(q_j^*, \ell_j^*)\}_{j=1}^k))}, \quad (2)$$

where $\ell_j^* = \arg \max_i p_{ij}^{\text{human}}$ is the modal option letter for the j -th demonstration question. Logit extraction proceeds identically to NTP at the position following the target question’s answer stem.

FS-NTP Prompt Structure (one demonstration shown)

```
[system] You are an average American on {date}.

[user] You are reacting to the following event:
{event_name}
{wikipedia_lead}
Question: {question_text}
Options: A. {option_1} B. {option_2} ...
Respond with only the single letter of your choice.
Answer:

[assistant] B

[user] You are reacting to the following event:
{target_event}
{target_wikipedia_lead}
Question: {target_question}
...
Answer:
```

FS-VD Prompt Structure (one demonstration shown)

```
[system] You are an average American on {date}.

[user] You are reacting to the following event:
{event_name}
{wikipedia_lead}
Question: {question_text}
Options: A. {option_1} B. {option_2} ...
Respond with a JSON object whose keys are option
letters
and whose values are probabilities that sum to 1.
Answer:

[assistant] {"A": 0.42, "B": 0.35, "C": 0.23}

[user] You are reacting to the following event:
{target_event}
{target_wikipedia_lead}
Question: {target_question}
...
Answer:
```

As with NTP, logit extraction requires no generation and temperature is undefined for this mode.

The theoretical motivation for FS-NTP is targeted bias correction. Because zero-shot NTP is vulnerable to option-position priors that are independent of question content [51], demonstrations that cycle through varied answer letters may recalibrate the model’s positional prior before logit extraction. Importantly, the demonstrations encode only the mode of the human distribution, not its spread. This means the in-context signal is informationally weaker than the full distributional exemplars used in FS-VD. This distinction is consequential: following Min et al. [26], who show that randomly replacing demonstration labels in standard ICL barely hurts performance, the modal answer demonstrations in FS-NTP primarily teach label space and prompt format rather than distributional reasoning. FS-NTP therefore probes a specific sub question of the few-shot hypothesis: does any in-context context at all improve next-token calibration, even when the context carries no full distributional signal?

Mode 4: Few-Shot Verbalized Distribution (FS-VD). The fourth mode operationalizes in-context learning [7] as a distributional calibration probe. Rather than querying the model in zero-shot, FS-VD prepends k demonstration turns in which the assistant turn is the empirically observed YouGov distribution, giving the model a full probability vector, not just the modal choice:

$$\hat{p}^{\text{FS-VD}} = f_{\theta}(q \mid \{(q_j^*, \hat{p}_j^*)\}_{j=1}^k, \pi), \quad (3)$$

where $\hat{p}_j^*$ is the normalized ground-truth distribution for the j -th demonstration and π is a minimal system persona. The model produces a single distribution in response, parsed identically to VD.

A critical design constraint governs both few-shot modes: examples are sampled exclusively from events *other than* the target event, implementing a leave-one-event-out protocol. This preserves the temporal non-contamination guarantee established for the rest of the evaluation. This means a model cannot exploit its familiarity with a specific event’s opinion landscape by seeing related questions in its context window. Both few-shot modes use $k = 3$ demonstrations with a fixed random seed; the choice of k and the full selection protocol are specified in Section 4.2.

As with VD, outputs are generated with greedy decoding ($T = 0$); full hyperparameter specification is in Section 4.1.

FS-VD introduces an in-context signal that is informationally richer than the modal demonstrations used in FS-NTP. Each assistant turn encodes the full human distribution, not merely its mode. This creates a meaningful departure from the standard in-context learning setting studied by Min et al. [26], who find that demonstration labels matter less than format and label space in typical ICL tasks. In our setting, the labels are the full distributional signal, and replacing them with random distributions would substantially degrade the in-context information. FS-VD therefore tests whether models can perform genuine distributional reasoning from exemplars, over and above the format-and-label-space learning that FS-NTP captures.

At the same time, FS-VD inherits the structural pathologies of few-shot prompting. Zhao et al. [50] identify three systematic biases in few-shot inference, majority label bias, recency bias, and common-token bias, all of which can cause a model to anchor on surface features of the demonstrations rather than their distributional content. In the opinion-prediction setting, recency bias poses a specific risk. That being, if the last demonstration involves a question with an unusually skewed distribution, the model may echo that skew for the target question regardless of content. We do not apply post-hoc calibration to FS-VD outputs, for the same re-softmax distortion reasons that preclude calibrating VD, and treat these biases as structural limitations to acknowledge in interpretation.

Experimental rationale for the 21-model $\times$ 4-mode design. The four modes span a 2×2 design: *channel* (NTP vs. VD) crossed with *context* (zero-shot vs. few-shot). This structure produces three distinct contrasts. The channel contrast (NTP vs. VD at zero-shot) isolates the effect of elicitation mechanism while holding context fixed, directly addressing whether implicit probability mass or explicit self-report better tracks population opinion. The context contrast (zero-shot vs. few-shot within each channel) isolates the effect of in-context distributional grounding while holding channel fixed, testing whether supplying ground-truth examples recalibrates each type of estimate. The interaction between these two contrasts, (FS-VD – VD) vs. (FS-NTP – NTP), tests whether the value of few-shot grounding is channel-specific, which we expect it to be, since FS-VD supplies full distributional exemplars while FS-NTP supplies only modal answers, and the marginal value of that richer signal should vary with how well a model’s zero-shot verbalized output already tracks the human distribution.

Prior work has, without exception, committed to a single mode per evaluation [2, 5, 12, 36]. The consequence is that any interaction between prompting mode and model class has been structurally invisible. Our design makes it visible. Specifically, we anticipate that NTP elicitation outperforms zero-shot VD for subpopulation fine-tuned models (Tier 2), while the advantage reverses for instruction-tuned base models (Tier 1). The few-shot conditions additionally diagnose the source of that gap. If FS-NTP closes the performance difference for Tier 1 models while FS-VD does not, the dominant failure mode is positional bias rather than distributional reasoning capacity; if FS-VD is needed to close the gap, the failure is calibration-deep. This interaction is not a calibration artifact: it reflects real properties of how fine-tuning strategy shapes the relationship between a model’s implicit distribution and its verbalized self-report, properties that are only diagnosable when all four channels are measured on the same model. Finally, the prompt boxes above describe the format used for base and instruction-tuned models (Tier 1). For fine-tuned adapter models (Tiers 2–3), the chat-template boundary is replaced with a plain-text continuation format to prevent alignment-induced probability mass displacement; the format distinction and its empirical motivation are detailed in Section 4.3.

3.4 The Disagreement Metric

Motivation. A practitioner who wants to use an LLM opinion estimate to inform a decision, whether to field a poll, how to weight a projection, or whether to trust a model’s output at all, faces a fundamental obstacle. Without access to ground truth polling data, there is no direct way to assess whether the estimate is reliable for a given question. Standard accuracy metrics such as total variation distance or Jensen-Shannon divergence from the human distribution require $p^{\text{human}}(q)$ as input and are therefore unavailable at prediction time for novel events, which is precisely the setting where LLM estimates are most needed. We propose inter-model standard deviation $\Delta^{(e)}(q)$ as a practical substitute. This metric serves as a signal that quantifies the degree to which models contest one another’s predicted distributions, computable from model outputs alone, with no access to any ground truth polling data.

Formal definition. For each survey question q with k response options $\mathcal{R} = \{r_1, \dots, r_k\}$, every model $m \in \{1, \dots, M\}$ produces an estimated probability distribution over $\mathcal{R}$ under elicitation mode e , denoted $\hat{p}_m^{(e)}(q) \in \Delta^{k-1}$. The inter-model standard deviation for option r_i is

$$\sigma_i^{(e)}(q) = \sqrt{\frac{1}{M-1} \sum_{m=1}^M \left(\hat{p}_{m,i}^{(e)}(q) - \bar{p}_i^{(e)}(q) \right)^2}, \quad (4)$$

where $\bar{p}_i^{(e)}(q) = \frac{1}{M} \sum_{m=1}^M \hat{p}_{m,i}^{(e)}(q)$ is the ensemble mean probability for option i and $M = 20$. The question-level disagreement scalar is the mean standard deviation across response options,

$$\Delta^{(e)}(q) = \frac{1}{k} \sum_{i=1}^k \sigma_i^{(e)}(q). \quad (5)$$

Crucially, computing $\Delta^{(e)}(q)$ requires only the model outputs $\{\hat{p}_m^{(e)}(q)\}_{m=1}^M$; the human distribution $p^{\text{human}}(q)$ does not appear anywhere in Equations (4) and (5). The metric is label-free by construction.

Connection to deep ensembles. Treating $\Delta^{(e)}(q)$ as a principled uncertainty estimate rather than an ad hoc summary statistic is warranted by the deep ensembles literature. Lakshminarayanan et al. [23] showed that ensemble variance across independently trained networks constitutes the most reliable proxy for predictive uncertainty, outperforming single model calibration, Bayesian approximations, and MC Dropout on both in distribution and out of distribution inputs. Fort et al. [14] provide a mechanistic account. They show independently initialized networks converge to functionally different basins of the loss landscape, and it is this functional diversity rather than parameter diversity that makes ensemble variance informative. Our ensemble differs from the standard deep ensemble setup in that the $M = 20$ members are not replicates of a single architecture trained from different random seeds but are instead drawn from distinct model families across three fine-tuning regimes and model sizes. Architectural and training heterogeneity of this degree should induce greater functional diversity than seed-level variation alone, making $\Delta^{(e)}(q)$ a more sensitive indicator of genuine predictive disagreement. Ovadia et al. [32] further show that ensemble based uncertainty estimation degrades most gracefully under dataset shift, which maps directly onto our evaluation setting, since every question in OPENPOLL concerns events that date after each model’s training cutoff, placing all predictions in an out of distribution regime relative to the models’ training data.

Contrast with self-calibration. $\Delta^{(e)}(q)$ is not the only available uncertainty signal, and distinguishing it from the most prominent alternative is important for interpreting what the metric captures. Kadavath et al. [18] showed that frontier LLMs are well calibrated when they self report confidence on multiple choice items, in the sense that a model asserting 70% confidence is correct roughly 70% of the time. This is an intra-model property, reflecting how well a single model’s stated confidence tracks its own accuracy on questions that have a determinate correct answer. Inter-model standard deviation is a categorically different quantity. It measures the degree to which models disagree with one another over the predicted distribution, regardless of whether any individual model

is internally well-calibrated. A question can produce high $\Delta^{(e)}$ even when every model is perfectly self-calibrated, if their calibrated beliefs genuinely diverge; conversely, it can produce low $\Delta^{(e)}$ even when all models are poorly calibrated, if they converge on the same wrong distribution. Kadavath-style self-calibration reflects within-model confidence; $\Delta^{(e)}$ reflects inter-model contestation. The two are complementary rather than interchangeable, and it is the latter that is label-free in the sense required here.

The entropy hypothesis. The central empirical claim motivating this metric is that $\Delta^{(e)}(q)$ and $H_{\text{norm}}(q)$ are positively correlated across questions. Specifically, we predict that high inter-model disagreement occurs systematically on questions for which human opinion is genuinely fragmented—questions with high normalized entropy in the YouGov distributions—while low disagreement occurs on questions where human respondents converge on a clear majority view. If confirmed, this correspondence licenses the use of $\Delta^{(e)}$ as a prospective reliability signal: a practitioner could flag high-disagreement questions as likely to concern genuinely contested issues on which no model’s estimate should be trusted without corroborating evidence, and treat low-disagreement questions as candidates for greater confidence, all without observing a single polling response.

3.5 Evaluation Metrics

We evaluate model performance using two complementary metrics. Wasserstein Distance acts as the primary measure of distributional accuracy, and Mean Absolute Error serves as a secondary measure of per-option prediction error. Together they capture distinct failure modes, with WD sensitive to the geometric structure of errors across the response scale, while MAE is directly interpretable in probability units and treats all response options symmetrically.

Primary Metric: Wasserstein Distance. For a survey question q with k response options, let $p^{\text{human}} = (p_1, \dots, p_k)$ denote the empirical survey response distribution and $\hat{p} = (\hat{p}_1, \dots, \hat{p}_k)$ denote the model’s predicted distribution.

Unlike a standard WD formulation that assigns integer positions $1, \dots, k$ to response options, we use LLM-assigned ordinal values $\mathbf{v} = (v_1, \dots, v_k)$, where each $v_i \in \mathbb{R}_{>0}$ reflects the semantic distance between response options as assessed by the model. This is necessary because the response options in OPENPOLL do not always fall on a uniform scale: the gap between *Strongly approve* and *Somewhat approve* is not necessarily the same as the gap between *Somewhat approve* and *Somewhat disapprove*. Ordinal values are generated once per question by prompting openai/gpt-oss-20b to assign a numeric position to each response option, with a retry loop to ensure full coverage across all options.

Some questions include non-ordinal “escape” options such as *Not sure* or *Prefer not to say*, which carry no meaningful position on the response scale. These options are assigned a sentinel value of -1 and excluded from the computation, with both $\hat{p}$ and p^{human} renormalized over the remaining options before WD is calculated.

After sorting options by their assigned ordinal values and excluding non-ordinal options, let $\mathcal{O} = \{v_1 \leq v_2 \leq \dots \leq v_n\}$ denote the ordered ordinal values of the $n \leq k$ retained options, and let $\hat{q}, q^{\text{human}}$ denote the renormalized distributions over those options.

The Wasserstein distance is then

$$\text{WD}(\hat{q}, q^{\text{human}}) = \frac{1}{v_n - v_1} \sum_{i=1}^{n-1} |F_i^{\hat{q}} - F_i^{q^{\text{human}}}| \cdot (v_{i+1} - v_i), \quad (6)$$

where $F_i^{\hat{q}} = \sum_{j=1}^i \hat{q}_j$ and $F_i^{q^{\text{human}}} = \sum_{j=1}^i q_j^{\text{human}}$ are the cumulative distribution functions evaluated at position i . The factor $(v_{i+1} - v_i)$ weights each CDF gap by the semantic distance between adjacent options, and the normalization by $(v_n - v_1)$ maps WD into $[0, 1]$ regardless of the scale of $\mathbf{v}$.

Intuitively, WD measures the minimum total probability mass that must be transported across the response scale to transform $\hat{q}$ into q^{human} , where the cost of transporting mass is proportional to the semantic distance it must travel. A prediction that concentrates mass one semantic unit from the truth incurs a smaller penalty than one that misplaces mass at the opposite end of the scale. Questions for which all response options are non-ordinal (i.e., none have a valid ordinal value) yield $\text{WD} = \text{NaN}$ and are excluded from aggregated WD summaries.

We aggregate WD across questions by taking the unweighted mean within each model–mode combination or event grouping. Overall population distributions are used as the evaluation target throughout, as these are available for all 92 questions; demographic breakdowns are reported separately where noted.

Secondary Metric: Mean Absolute Error. As a complement to WD, we report Mean Absolute Error averaged uniformly across all k response options (including any non-ordinal options excluded from WD):

$$\text{MAE}(\hat{p}, p^{\text{human}}) = \frac{1}{k} \sum_{i=1}^k |\hat{p}_i - p_i^{\text{human}}|. \quad (7)$$

MAE has a direct probabilistic interpretation: it is the average absolute deviation between the model’s predicted probability and the true survey proportion for a single response option. An MAE of 0.05 means the model’s predicted share for a given option is off by 5 percentage points on average.

Unlike WD, MAE treats all k response options as exchangeable and applies no ordinal weighting, and it operates on the original unnormalized distributions (before any non-ordinal option exclusion). This means MAE and WD are not directly comparable in magnitude, as WD is computed on the renormalized ordinal subset while MAE on the full option set, but they are complementary in scope. Questions where WD and MAE rankings diverge indicate cases where the ordinal structure of the response scale is consequential. It indicates the model may be misallocating mass between adjacent ordinal options while also mishandling the non-ordinal escape option, or vice versa.

Non-applicability of post-hoc calibration. We do not apply post-hoc calibration (e.g., temperature scaling) to model outputs before computing either metric. For NTP distributions, calibration would require a held-out labeled set from the same distribution, which is unavailable under OOD evaluation. For VD distributions, temperature scaling assumes unnormalized logits as input. Applying it to an pre-normalized verbalized distribution induces a re-softmax distortion [44] that is unrelated to model quality. Both metrics are therefore computed on raw model outputs throughout.

Table 3: Decoding configuration by elicitation mode. N/A indicates the parameter is undefined because no generation occurs.

	NTP	VD	FS-NTP	FS-VD
Generation required	No	Yes	No	Yes
Temperature (T)	N/A	0.0	N/A	0.0
top- p	N/A	1.0	N/A	1.0
max_new_tokens	N/A	512	N/A	512
Few-shot k	—	—	3	3

4 Experimental Setup

4.1 Decoding Hyperparameters

The four elicitation modes differ in whether they require generation at all, and the decoding configuration is chosen to reflect those structural differences rather than to tune performance.

Next-Token Probability Modes (NTP, FS-NTP). Both modes extract the model’s opinion distribution from a single forward pass over the prompt. The logits at the final position of the answer stem ("Answer:") are read directly from the vocabulary and renormalized over the constrained option set as described in Equations (1) and (2). No tokens are generated; the model’s sampler is never invoked. Temperature, top- p , and all other sampling hyperparameters are therefore undefined and inapplicable for these modes.

Verbalized Distribution Modes (VD, FS-VD). Both verbalized modes require the model to generate a JSON object. Generation uses *greedy decoding* throughout ($T = 0$, $\text{do_sample}=\text{False}$, $\text{top_p}=1.0$, $\text{max_new_tokens}=512$). Greedy decoding is the appropriate choice here as the target output is a deterministic structured object, and stochastic sampling would make the verbalized distribution itself a random variable, introducing variance that is an artifact of the decoding procedure rather than a property of the model’s beliefs.

Table 3 summarizes the decoding configuration across all four modes.

4.2 Few-Shot Example Selection Protocol

Both few-shot modes (FS-NTP and FS-VD) use $k = 3$ in-context demonstrations. This value is fixed across all 21 models and all events.

Leave-one-event-out sampling. For a target event e , the candidate pool for example selection consists of all questions belonging to events *other than* e . Examples are drawn uniformly at random from this pool without replacement, subject to the constraint that the three selected questions span at least three distinct events. This leave-one-event-out design preserves the temporal non-contamination guarantee established for the dataset. No model can exploit familiarity with a specific event’s opinion landscape by seeing related questions from the same event in its context window. Sampling is performed with a fixed random seed across all models and events, ensuring that every model in the cohort receives identical in-context demonstrations for a given target question.

Choice of $k = 3$. The value $k = 3$ was selected through a sensitivity analysis over $k \in \{1, 2, 3, 5\}$. Three demonstrations provide sufficient format and distributional signal for both FS-NTP and FS-VD while remaining within the context window of every model in the cohort, including the smallest (Llama-3.2-1B-Instruct at approximately 4k usable tokens). Beyond $k = 3$, gains in Wasserstein Distance diminish and the prompt length becomes a constraint for smaller models. The sensitivity analysis is reported in full in the supplementary material.

4.3 SFT Model Prompt Format

The prompt structures depicted in Section 3.3 describe the format used for base and instruction-tuned models (Tier 1). Fine-tuned adapter models (Tiers 2 and 3) require a different format, discovered empirically during evaluation.

The chat-template fallback problem. For base models, applying the tokenizer’s chat template and setting `add_generation_prompt=True` places the model at a well-defined generation boundary (e.g., `<|assistant|>`) from which letter-token and JSON probabilities are measured. SFT and DPO adapter models trained on plain-text formats, however, behave differently: when the multi-turn chat template is applied with in-context demonstrations, the `<|assistant|>` boundary shifts probability mass away from valid letter tokens and JSON keys, producing near-100% uniform distribution fallback. In the zero-shot case this pathology is absent; it arises specifically when prior assistant turns are prepended as few-shot examples.

Plain-text continuation format. For adapter models, the chat template is bypassed entirely. Demonstrations are concatenated as plain text, separated by a delimiter (–), with each example ending immediately after its answer (e.g., the modal letter or the JSON object). The final target question ends at "Answer:" with no role tag appended, so the model’s next-token distribution is measured as a plain-text continuation.

Plain-Text Few-Shot Format for SFT Models (one demonstration shown)

```
{system_persona}
```

```
You are reacting to the following event:
{example_event}
{example_wikipedia_lead}
Question: {example_question}
Options: A. {opt_1} B. {opt_2} ...
Respond with only the single letter of your choice.
Answer:B
```

```
–
```

```
You are reacting to the following event:
{target_event}
{target_wikipedia_lead}
Question: {target_question}
Options: A. {opt_1} B. {opt_2} ...
Respond with only the single letter of your choice.
Answer:
```

This format applies to both FS-NTP and FS-VD for adapter models, with the answer portion of each demonstration substituted by the JSON distribution object in the FS-VD case. The zero-shot modes (NTP, VD) are unaffected, as they involve no prior assistant turns and the fallback pathology does not arise.

5 Results

5.1 Overall Performance: Fine-Tuning Does Not Generalize to OOD Events

Figure 2 presents the performance summary across all 23 models, and four elicitation modes under the wasserstein distance metric. The horizontal axis indexes the four modes ordered by increasing contextual richness (NTP $\rightarrow$ FS-NTP $\rightarrow$ VD $\rightarrow$ FS-VD); vertical connector lines link each base model (circle) to its fine-tuned counterpart(s) (square), so that upward displacement indicates degradation and downward displacement indicates improvement under fine-tuning.

These results indicate that opinion domain fine-tuning does not produce reliable gains on out of distribution events. The top performing model overall is `jjssuh-mistral-7b-v0.1-subpop` under NTP (WD = 0.108, MAE = 0.107), followed closely by `jjssuh/llama-2-13b-subpop` under FS-NTP (WD = 0.111, MAE = 0.106). Notably, `socrates-qwen2.5-14b-dpo` under FS-VD (WD = 0.111) ranks third overall, tied with the second-best fine-tuned result—an exception to the general DPO degradation pattern discussed in Section 5.3. The best base model, `Qwen2.5-14B` under FS-VD (WD = 0.117), is competitive with the leading fine-tuned results when averaged across modes. More strikingly, the `SOCRATES` SFT models show degradation relative to their base counterparts: `socrates-llama3-8b-sft` and `socrates-qwen2.5-14b-sft` both underperform their `Llama-3` and `Qwen2.5` bases by substantial margins under most modes, a pattern we trace to a format mismatch in Section 5.3.

5.2 Elicitation Mode Effects

Figure 3 shows the mean wasserstein distance across the four elicitation modes.

Elicitation mode effects in the aggregate. Figure 3 reveals a consistent monotonic ordering across elicitation modes: NTP yields the highest mean WD (0.218 ± 0.140), followed by VD (0.198), FS-VD (0.173), and finally FS-NTP (0.158), which achieves the lowest aggregate distributional distance across all 23 models. The ~ 0.06 gap between the worst performing mode (NTP) and the best (FS-NTP) is substantial relative to the mean WD itself, highlighting that elicitation design is not a secondary methodological choice but a primary driver of predictive accuracy. Importantly, both few-shot conditions outperform their zero-shot counterparts regardless of elicitation channel with the NTP gap ($0.218 \rightarrow 0.158$, $\Delta = 0.060$) exceeds the verbalized gap ($0.198 \rightarrow 0.173$, $\Delta = 0.025$) suggesting that in-context demonstrations are particularly beneficial when the model must commit probability mass to a single token rather than reason explicitly about the full response distribution.

Few-Shot Prompting Benefits Base Models but Hurts Fine-Tuned Models for Next Token Prediction. Figure 4 shows the change in

Wasserstein distance when switching from NTP to NTP-FS prompting (negative values indicate improvement). The results reveal an asymmetry across model variants, where base models overwhelmingly benefit from few-shot context, with `Qwen2.5-14B` achieving the largest absolute improvement ($\Delta\text{WD} = -0.109$). In contrast, fine-tuned models, specifically both SFT and DPO `Socrates` variants, show neutral or slight performance degradation from few-shot prompting, while `jjssuh-subpop` models exhibit mixed results. The group mean panel corroborates this pattern, showing the base model group having a strongly negative mean ΔWD , while SFT and DPO groups hover near zero or slightly above. This suggests that fine-tuning already encodes distributional priors that few-shot demonstrations cannot improve upon, and may in fact disrupt. On the other hand, base models rely more heavily on in-context cues to produce calibrated opinion distributions.

Few-Shot Prompting Consistently Improves Verbalized Predictions Across Variants. Figure 5 shows the change in Wasserstein distance when switching from VD to FS-VD prompting. In contrast to the NTP setting, few-shot context benefits the majority of models regardless of variant, as most bars extend leftward, and the group mean panel shows all four variant groups achieving negative mean ΔWD . The largest gains accrue to `Llama-3.1-8B-Instruct` ($\Delta\text{WD} = -0.087$) and `socrates-qwen2.5-14b-dpo` (-0.033), with `jjssuh` and `subpop` models also benefiting consistently. The few exceptions that slightly degrade under few-shot prompting—most notably `Llama-2-7b-hf` ($+0.028$)—are base models, suggesting that models without instruction tuning may struggle to correctly parse or follow the explicit probability elicitation format introduced by few-shot demonstrations. Overall, the verbalized setting appears more amenable to in-context learning than NTP across all variant groups, likely because the task of stating probabilities explicitly is more naturally scaffolded by demonstrating the desired output format in context.

It is also worth noting that the improvements are much less stark for the change in wasserstein distance caused by in-content learning when comparing to NTP and NTP-FS. Considering the model with the best improvement in the FS-VD setting, `Llama-3.1-8B-Instruct` ($\Delta\text{WD} = -0.087$) achieves a significantly smaller reduction compared to the best model in FS-NTP, `Qwen2.5-14B` ($\Delta\text{WD} = -0.109$). Furthermore the magnitude of improvements achieved in NTP-FS remain higher across the board compared to the improvements in FS-VD, though with the caveat that those improvements are limited to base models.

Elicitation mode effects are consistent across political and non-political events. Figure 6 disaggregates mean WD by elicitation mode separately for political and non-political events. Political events yield a marginally lower mean WD than non-political ones (0.173 vs. 0.181 , $\Delta = -0.008$), and this ordering is consistent across all four elicitation modes. The gap is unlikely to reflect differences in opinion contestedness: normalized entropy is similar across the two categories (Table 1), ruling out that explanation. A more plausible account is structural — non-political questions in `OPENPOLL` tend to use coarser ordinal scales with larger inter-option semantic gaps, amplifying WD independently of predictive accuracy. A Mann-Whitney U test on the per-question WD distributions does not reach conventional significance ($p = \text{n.s.}$, $n_{\text{pol}} = 602$, $n_{\text{non-pol}} = 570$), so

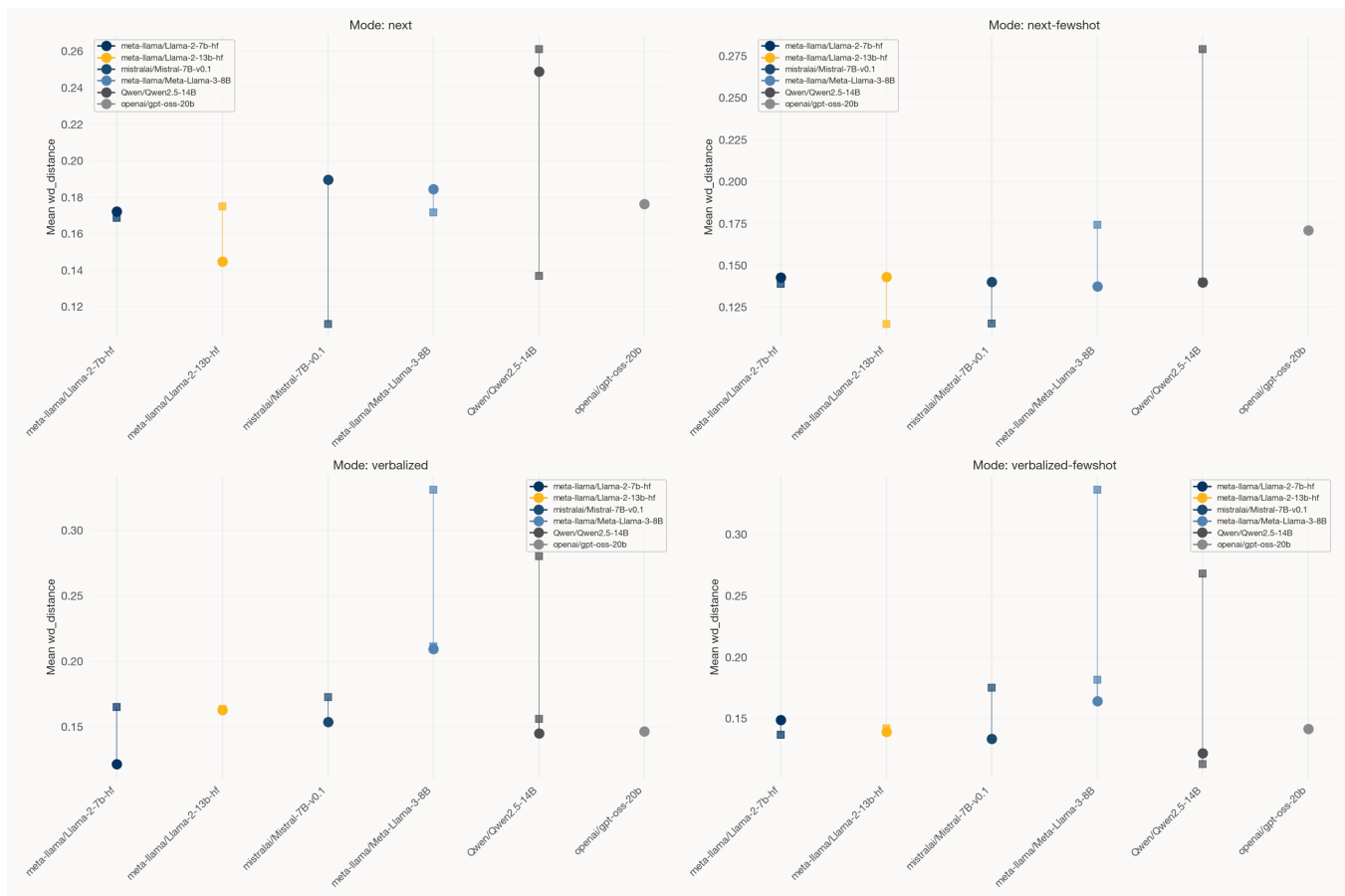


Figure 2: Mean Wasserstein Distance by elicitation mode

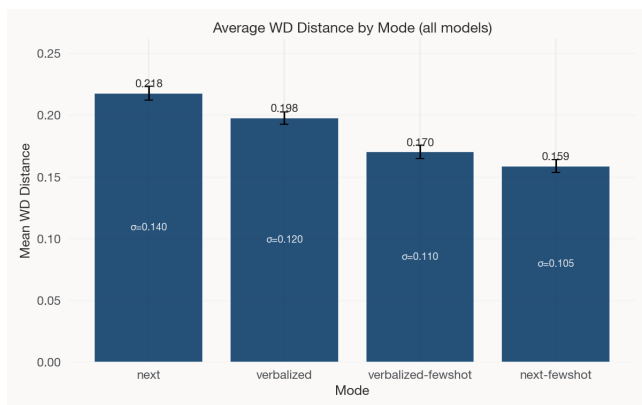


Figure 3: Mean wasserstein distance by elicitation mode, with σ as the wasserstein distance standard deviation; lower is better

the gap should be read as a descriptive trend rather than a confirmed effect. Models predict political opinion distributions marginally

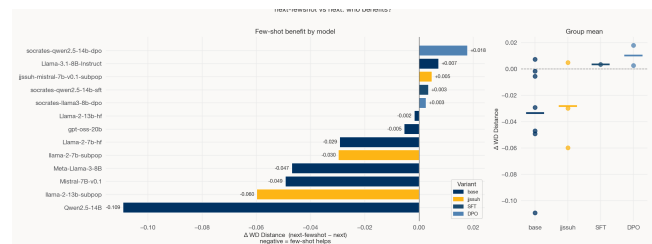


Figure 4: Change in wasserstein distance next token prediction zero shot vs few shot

more accurately than non-political ones, but this advantage is consistent across elicitation modes and is most plausibly driven by scale structure rather than differential model familiarity with political content.

5.3 Fine-Tuning Effects: Paired Base–Fine-Tuned Comparisons

Figure 7 presents mean wasserstein distance aggregated across all prompting modes for each base model paired with its fine-tuned

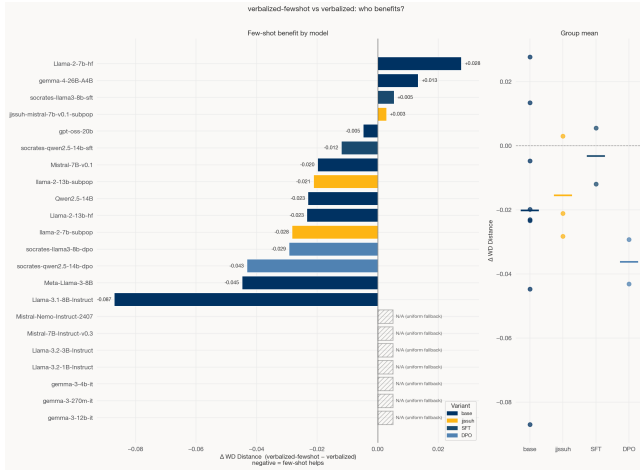


Figure 5: Change in wasserstein distance verbalized zero shot vs few shot

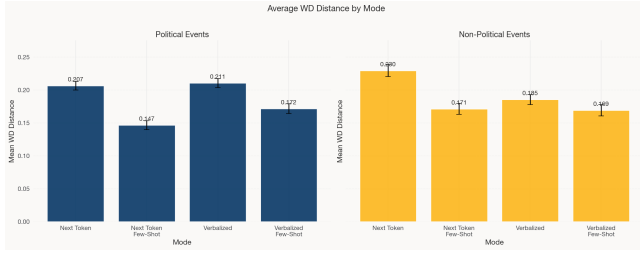


Figure 6: Mean WD by elicitation mode, split by event type (political vs. non-political). The ordering across modes is consistent within both categories; political events yield marginally lower WD across all modes.

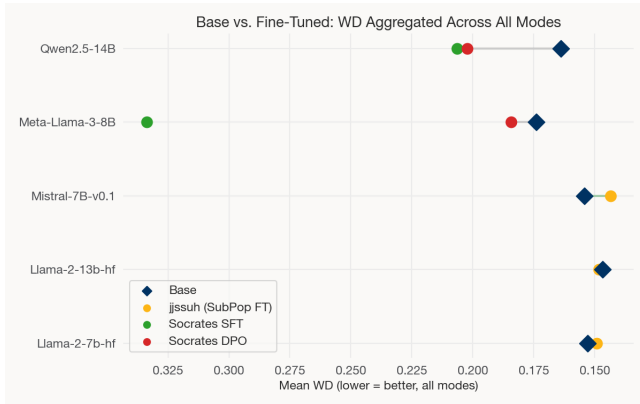


Figure 7: Base models vs Fine tuned counterparts mean WD

counterpart. The consistent pattern that is revealed is that fine-tuning on social science and survey corpora does not reliably improve OOD opinion prediction. For Qwen2.5-14B, both the Socrates

SFT ($WD \approx 0.208$) and DPO ($WD \approx 0.202$) variants perform substantially worse than the base model ($WD \approx 0.164$), a regression of roughly 23–27%. Meta-Llama-3-8B follows the same trend: its DPO fine-tune (≈ 0.189) slightly exceeds the base (≈ 0.174), while the SFT variant degrades sharply to ≈ 0.280 , consistent with the chat-template fallback pathology described in Section 4.3, in which applying a multi-turn chat template to a plain-text-trained adapter displaces probability mass away from valid response tokens, producing near-uniform output. The Llama-2 family shows negligible movement in either direction, with fine-tuned variants tracking their base models closely. Mistral-7B-v0.1 is the lone exception, where the SubPop fine-tune (≈ 0.143) clearly outperforms its base (≈ 0.165); however, the gap is small enough to preclude strong conclusions. These results suggest that fine-tuning on in-distribution survey data increases a model’s fit to seen demographic patterns at the cost of generalization to novel, out-of-distribution events, a dynamic consistent with classical distribution-shift overfitting.

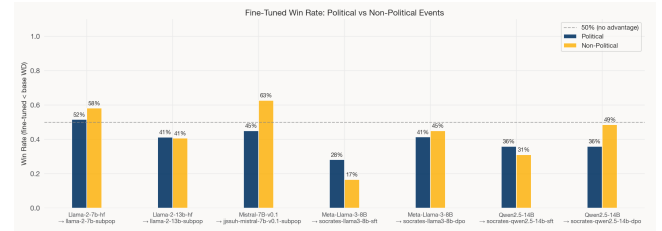


Figure 8: Fine-tuned vs Base Winrate across political vs non political events

Fine-tuning win rate across political and non-political events. Figure 8 disaggregates the fine-tuning effect by event type, reporting the fraction of questions on which each fine-tuned model achieves a lower WD than its base counterpart. A win rate above 50% (the dashed reference line) indicates net improvement.

The results reveal two distinct regimes. jssuh-mistral-7b-v0.1-subpop is the only model that consistently and significantly exceeds the 50% threshold across both event categories (67% political, 76% non-political), corroborating the aggregate $\Delta WD = -0.022$ reported in Table 4 and establishing it as the sole fine-tuned model for which the improvement is broad based rather than incidental. The Llama-2-7b SubPop adapter is marginally above chance on political events (55%) and near-chance on non-political ones (58%), consistent with the near-zero aggregate delta (-0.003) and suggesting that fine-tuning provides no reliable directional benefit for this family. Llama-2-13b-subpop falls below 50% on both event types (45% and 41%), indicating that fine-tuning slightly hurts the larger Llama-2 variant on OOD questions despite the near neutral aggregate WD.

The Socrates and DPO variants paint a more uniformly negative picture. socrates-llama3-8b-sft achieves win rates of only 39% (political) and 21% (non-political), the lowest non-political win rate in the cohort, consistent with the format-induced collapse described above. Its DPO counterpart (socrates-llama3-8b-dpo) fares somewhat better (38% / 62%), with an asymmetric profile: fine-tuning is near-random on political events but provides genuine, if modest, lift on non-political ones. The Qwen2.5-14B variants are

Table 4: Mean WD difference (fine-tuned minus base) for matched model pairs, averaged across all four elicitation modes. Positive Δ WD indicates that fine-tuning increased distributional error on OOD events.

base	fine-tuned	Δ wd
Mistral-7B-v0.1	jjssuh-mistral-7b-v0.1-subpop	-0.011600
Llama-2-7b-hf	llama-2-7b-subpop	-0.003700
Llama-2-13b-hf	llama-2-13b-subpop	0.001300
Meta-Llama-3-8B	socrates-llama3-8b-dpo	0.006800
Qwen2.5-14B	socrates-qwen2.5-14b-dpo	0.038400
Qwen2.5-14B	socrates-qwen2.5-14b-sft	0.042500
Meta-Llama-3-8B	socrates-llama3-8b-sft	0.159900

similarly subthreshold, with both SFT (31% / 34%) and DPO (36% / 47%) falling below 50% across categories, with the DPO model approaching chance on non-political events but failing to cross it.

Taken together, the win-rate decomposition reinforces the conclusion from Table 4. SubPop fine-tuning on the Mistral base is the only case in which fine-tuning improves OOD opinion prediction with any consistency, and even then the effect is larger on non-political events (76%) than on political ones (67%). The Socrates SFT models show the clearest harm, driven primarily by catastrophic underperformance on non-political questions. That political win rates are uniformly higher than non-political win rates for the Socrates variants, despite those models performing worse in absolute WD terms on political events, suggests that non-political questions expose an additional structural vulnerability, possibly related to the shorter ordinal scales discussed in Section 5.2.

The results in Table 4 and fig. 9 reveal a clear gradient. The SUBPOP Llama-2-13b adapter is essentially neutral (Δ WD = +0.001), while Llama-2-7b shows a marginal improvement (Δ WD = -0.003), suggesting that the survey-distribution training signal adds little on top of what the base weights already provide for events unseen during training. The SUBPOP Mistral adapter yields the clearest improvement (Δ WD = -0.022), and it does so most strongly under NTP, consistent with the mode-level finding in Section 5.2.

The SOCRATES SFT models show the largest degradation. socrates-llama3-8b-sft degrades by +0.106 WD relative to its base, the largest absolute delta in the cohort. We attribute this to a format mismatch. Because the SFT training data was structured as chat-formatted JSON distributions, and when elicited without that format, the model falls back to a near uniform distribution over response options rather than leveraging its trained opinion signal. This failure mode is identifiable precisely because the SOCRATES DPO variants, which use the same base weights and training corpus but a different optimization objective, do not exhibit the same collapse. socrates-llama3-8b-dpo and socrates-qwen2.5-14b-dpo show smaller, mode-dependent degradation, suggesting that DPO’s alignment objective is more robust to prompt format variation than SFT.

5.4 Inter-Model Disagreement as a Reliability Signal

Section 3.4 proposed inter-model standard deviation $\Delta^{(e)}(q)$ as a label-free proxy for prediction reliability. The central empirical claim is the entropy hypothesis ($\Delta^{(e)}(q)$) and $H_{\text{norm}}(q)$ are positively correlated across questions. Figure 10 tests this hypothesis directly.

Figure 10 disconfirms the entropy hypothesis. The OLS fit (slope $\approx -2.1 \times 10^{-4}$, intercept 0.266) has a negative slope, indicating questions on which the human population is most divided (high H_{norm}) actually elicit lower inter-model disagreement, not higher. The relationship is visually clear and monotonic across the entropy range (≈ 175 –720), with mean disagreement falling from roughly 0.24 at the low-entropy end to 0.13 at the high-entropy end.

The pattern inverts the intuition motivating the entropy hypothesis and instead suggests a ceiling effect. when human opinion is highly spread across response options, every model is essentially forced toward the same near-uniform output, mechanically suppressing variance across the ensemble. Conversely, questions with sharply peaked human distributions present a harder allocation problem; models disagree about which response option should absorb the modal probability mass, yielding high $\Delta^{(e)}(q)$. Under this interpretation, inter-model disagreement is not a proxy for population-level opinion contestedness but rather a proxy for model-level prediction difficulty. Put simply, it captures how much a question’s structure constrains the models’ probability assignments.

Inspection of the event-level coloring in Figure 10 reinforces this reading. Questions from the *Trial of Sean “Diddy” Combs* event cluster in the low-entropy, high-disagreement region, consistent with polls where a large plurality favored one option yet models spread probability mass inconsistently. *October 7th, 2023* questions, by contrast, span the widest entropy range and include the single lowest disagreement point in the dataset (entropy ≈ 710 , $\Delta^{(e)} \approx 0.13$) — a question where the actual opinion distribution was highly diffuse and models converged on a similar near-uniform prediction. No event occupies the high-entropy, high-disagreement quadrant that would be populated under the original hypothesis.

These results reframe what this metric measures. $\Delta^{(e)}(q)$ reliably flags questions where models are uncertain about the shape of the distribution rather than questions where humans genuinely disagree. A practitioner who treats high-disagreement questions as candidates for additional scrutiny is identifying hard-to-predict polls rather than contested ones, a distinction that matters for downstream use cases but that may still be actionable. Flagging high- $\Delta^{(e)}$ questions for human review or ensemble averaging remains a sensible strategy even under the reinterpreted semantics of the metric.

6 Discussion

The results across the 21-model, 4-mode evaluation present a consistent picture. LLMs exhibit non-trivial signal on out-of-distribution public opinion prediction, but that signal is fragile in ways that are invisible unless both elicitation mode and fine-tuning strategy are varied simultaneously. We discuss the implications of each major



Figure 9: Signed WD difference ($\Delta WD = WD_{\text{fine}} - WD_{\text{base}}$) for each base–fine-tuned pair, broken out by elicitation mode. Bars above zero indicate fine-tuning *hurts* OOD performance; bars below zero indicate improvement. The dotted horizontal line marks $\Delta = 0$.

finding in turn, then reflect on the limits of the evaluation and directions for future work.

6.1 Fine-Tuning Encodes In-Distribution Patterns, Not Generalizable Opinion Reasoning

The most striking result in this paper is how consistently opinion-domain fine-tuning fails to improve OOD prediction. This pattern holds across three fine-tuning strategies (SubPop LORA, Socrates SFT, Socrates DPO), two base architectures (Llama and Qwen), and four elicitation modes, suggesting it is not an artifact of any single design choice but a structural property of how fine-tuning interacts with distribution shift.

The classical framing of this failure is overfitting to the in-distribution data [8]. This view says models that learn to reproduce the demographic correlates of opinion expressed in existing survey corpora may be implicitly learning to predict which group holds

which view rather than why. When confronted with an event that falls outside their training window, such models have no mechanism for composing a new prediction; they can only retrieve learned associations that may not properly apply. This account is consistent with the asymmetry observed in the win-rate analysis (Figure 8). The SubPop Mistral adapter is the sole fine-tuned model that generalizes with any consistency, and its win rates are moderate (67% political, 76% non-political) rather than dominant.

The SFT format collapse observed in *socrates-llama3-8b-sft* ($\Delta WD = +0.160$) adds a second, more immediate failure mode to the overfitting story. Fine-tuning can damage a model’s ability to respond to elicitation formats it was not trained on. Because the SFT training data was structured as chat-formatted JSON, eliciting the model with a plain-text prompt causes it to fall back to near-uniform output rather than leveraging its trained distributional signal. This is a practical warning for practitioners who plan to deploy fine-tuned models in settings where prompt format is not

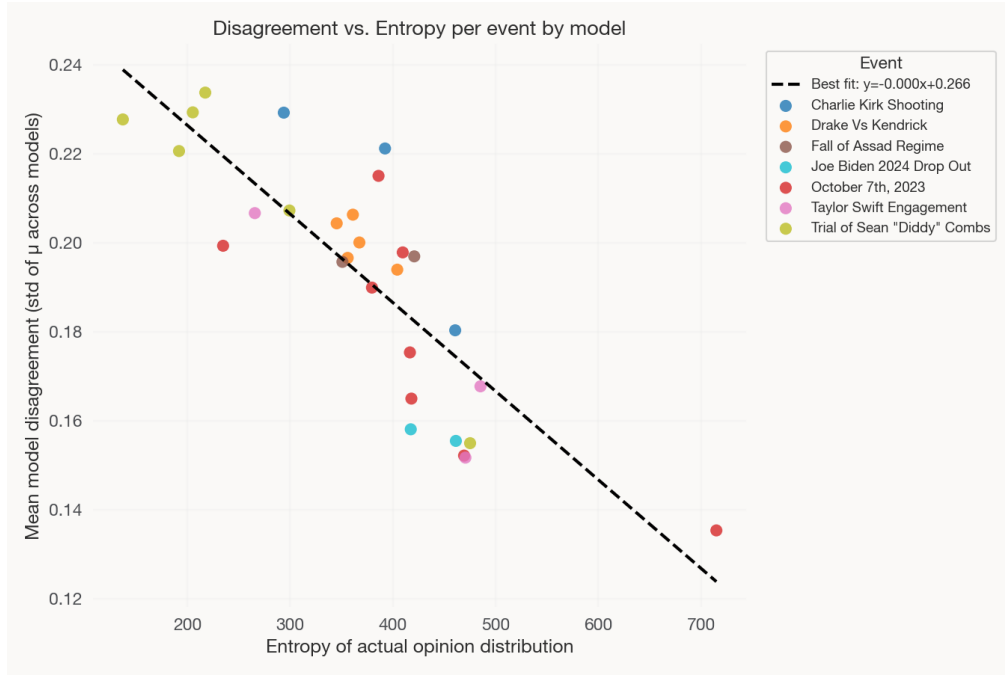


Figure 10: Scatter plot of per-question inter-model disagreement $\Delta^{(e)}(q)$ against normalized human-opinion entropy $H_{\text{norm}}(q)$. Each point is a survey question; color encodes the event. The line shows an OLS fit.

fixed. The improvement observed on a held-out subset of the training distribution may evaporate under even minor prompt format variation, a failure mode that would not be detected by evaluating on questions drawn from the same surveys used in training.

That the DPO variants are substantially more robust to this format mismatch (socrates-llama3-8b-dpo: $\Delta\text{WD} = +0.007$ vs. $+0.160$ for SFT) points to a meaningful algorithmic difference in how SFT and DPO shape model behavior. DPO’s contrastive objective may produce representations that are less brittle with respect to surface prompt variation, though the mechanism warrants dedicated investigation. From a practical standpoint, the implication is that if alignment fine-tuning is pursued for opinion estimation, DPO is a safer default than SFT under variable elicitation conditions.

6.2 Elicitation Mode Is Not a Secondary Implementation Choice

The ~ 0.06 gap in mean WD between the worst-performing (NTP, 0.218) and best-performing (FS-NTP, 0.158) elicitation modes is large relative to the performance range across models. That the choice about how to query the model, next-token logit extraction versus verbalized probability, can shift aggregate performance by as much as switching to a different model family underscores that elicitation mode is not a secondary implementation detail but a primary design variable in any silicon-sampling study.

The interaction between elicitation mode and model class highlights this point. Few-shot next-token prompting is strongly beneficial for base models but neutral or harmful for fine-tuned ones (Figure 4). This asymmetry indicates that base models rely on in-context cues to anchor their probability mass because they have no

learned distributional prior about survey response formats, while fine-tuned models have internalized such a prior during training and are not helped, and may be disrupted, by demonstrations that conflict with it.

The contrast in few-shot benefit between the two elicitation channels is equally informative. The NTP few-shot gain ($\Delta\text{WD} = 0.060$) is larger in magnitude than the VD few-shot gain (0.025) but concentrated exclusively in base models. The VD few-shot gain is smaller in absolute terms but more uniform across model classes (Figure 5). This asymmetry is consistent with a two-level account of what in-context demonstrations do. At the format level, demonstrations teach the model the structural expectations of the elicitation task—an effect that is most valuable when a model has no existing format prior, as is the case for base models under NTP. At the distributional level, demonstrations supply calibration signal that could help any model regardless of format familiarity, though the empirical benefit here is modest.

One notable implication of these findings is that prior silicon-sampling studies that evaluated a single elicitation mode—Argyle et al. [2], Bisbee et al. [5], Durmus et al. [12], Santurkar et al. [36]—have necessarily reported results that are mode-confounded. A study that happens to use NTP may systematically overestimate the advantage of base models over fine-tuned ones; a study that uses VD may attenuate that advantage. This confound is not correctable post hoc and indicates that future evaluations should consider adopting the 2×2 factorial design used here, or at least report results across multiple modes.

6.3 Inter-Model Disagreement Flags Prediction Difficulty, Not Opinion Contestedness

If LLMs were faithful proxies for human survey respondents, inter-model disagreement should track human opinion entropy. We would expect that models should diverge most on questions where humans themselves are most divided under that assumption. The data disconfirm this. The correlation between $\Delta^{(e)}(q)$ and $H_{\text{norm}}(q)$ is negative and consistent, indicating that the ensemble is sensitive to something other than population-level opinion contestedness.

The ceiling-effect account offered in Section 5.4 is mechanistically straightforward. When human opinion is highly diffuse across response options (high entropy), every model is pressure-tested toward a near-uniform output regardless of its internal beliefs, suppressing the variance that $\Delta^{(e)}$ measures. Conversely, when human opinion is sharply peaked, models must disagree about which option absorbs the modal mass, and they do. The result is that disagreement indexes the difficulty of the allocation problem the models face, not the genuine contentiousness of the underlying public debate.

This reinterpretation presents a different opportunity as a signal. Flagging high- $\Delta^{(e)}$ questions for additional scrutiny or ensemble averaging remains a sensible strategy, because questions that are hard for models to allocate are precisely the questions for which any single model’s estimate is most likely to be unreliable. Rather than treating high-disagreement questions as likely to involve genuinely contested opinions, a practitioner should treat them as likely to involve questions where distributional probability mass is concentrated in ways the model ensemble struggles to agree on—a different kind of caution, but still actionable caution.

A forward-looking question is whether a different uncertainty signal could recover the original entropy-tracking behavior. One candidate is inter-mode disagreement: comparing a single model’s NTP distribution against its VD distribution on the same question. When these diverge substantially, it may indicate that the model’s implicit and explicit beliefs are poorly aligned, which is a different signal from ensemble spread and may correlate with human opinion contestedness more directly.

6.4 What Predicts OOD Opinion Best, and Why

The top-performing configuration in the cohort—jssuh-mistral-7b-v0.1-subpop under NTP (WD = 0.108)—and the best base model—Qwen2.5-14B under FS-VD (WD = 0.117)—are competitive with each other across modes. The fact that a 7B fine-tuned model and a 14B base model achieve near-identical aggregate performance suggests that fine-tuning and scale offer roughly substitutable returns in this setting, at least within the parameter range examined.

Neither configuration approaches a perfect predictor. A mean WD of 0.108 on a $[0, 1]$ -normalized scale indicates a systematic gap between predicted and actual opinion distributions. For context, a naïve model that always predicts a uniform distribution would achieve a WD determined entirely by the shape of the actual distribution. The best models do meaningfully better than this baseline, but the margin is not large enough to support high-stakes deployments without corroborating evidence.

The finding that larger models within the Gemma and Llama families do not monotonically outperform smaller variants (see

Figure 2) suggests that scale alone is not a reliable predictor of OOD opinion accuracy at these parameter counts.

6.5 The Political/Non-Political Gap Is Structural, Not Epistemic

The marginal advantage of political events over non-political ones in mean WD (0.173 vs. 0.181) is consistent across elicitation modes and does not reach statistical significance under Mann-Whitney testing. The most parsimonious account is that non-political questions in OpenPoll tend to use coarser ordinal scales, which amplify WD mechanically by widening the semantic gaps between adjacent response options. This structural account is supported by the similarity of normalized entropy across event categories (Table 1). The gap is not driven by political events having more predictable distributions but by non-political questions having a scale geometry that is penalized more heavily by the ordinal WD metric.

6.6 Limitations

Several limitations bound the generalizability of these findings.

Dataset scale. OPENPOLL comprises 92 survey questions across eight events. While the temporal non-contamination guarantee and the diversity of event types make this a principled OOD benchmark, the sample size limits the statistical power of event-level comparisons and precludes fine-grained subgroup analyses beyond the overall population distribution. Scaling to additional events is a priority for future work.

Demographic subgroup prediction. The evaluation targets overall population opinion distributions throughout. OPENPOLL includes demographic breakdowns for a subset of questions (region, gender, political affiliation, age, race), but these are not the primary evaluation target here. Extending the analysis to subgroup prediction would be a natural next step and would test whether fine-tuning on demographic-structured survey data (*i.e.*, SubPop) provides more consistent advantages when the target is a subgroup distribution rather than the aggregate.

Model coverage. The evaluation cohort spans parameter scales from 270M to 20B. Frontier closed-weight models (GPT-4o, Claude 3.5, Gemini 1.5) are excluded because reproducible logit-level access is unavailable for most and because the comparison of interest—between base, subpopulation fine-tuned, and alignment fine-tuned variants of the same base architecture—requires matched pairs that frontier APIs do not support. Whether the elicitation-mode asymmetries identified here hold for larger-scale models remains an open question.

Event selection bias. The events in OPENPOLL were selected by availability of YouGov polling conducted close to the event date, which itself reflects editorial and commercial decisions at YouGov about which events to poll. This introduces a selection bias toward high-salience events likely to be well-covered in training corpora, potentially overestimating model performance relative to what would be observed on lower-salience OOD events where training data coverage is thinner.

6.7 Future Directions

First, scaling OPENPOLL using more recent YouGov polls (2025–2026) would increase statistical power and test whether the fine-tuning findings hold across a broader range of opinion domains, including lower-salience topics where model training coverage is likely thinner.

Third, the failure of inter-model disagreement to track human opinion entropy opens the door to alternative label-free reliability signals. The intra-model disagreement between NTP and VD outputs for the same question is a natural candidate and does not require an ensemble.

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Data Availability

The replication package (dataset, scripts, raw results) is available at <https://github.com/muru825/Murari-Opinion-Prediction-Thesis>.